

Reliability Models: Logarithmic Poisson Model

Software Reliability Growth Model
(Musa-Okumoto Model)

Introduction

- The Logarithmic Poisson Model (LPM) is a software reliability growth model (SRGM) that describes how the number of software failures decreases over time as bugs are detected and fixed.
- • Proposed by Musa and Okumoto (1984)
- • Based on a Non-Homogeneous Poisson Process (NHPP)
- • Reliability growth slows logarithmically with testing time

Model Equations

- Expected cumulative failures:
- $\mu(t) = (1/\theta) \ln(1 + \theta\varphi t)$
- Failure intensity:
- $\lambda(t) = \varphi / (1 + \theta\varphi t)$
- Where:
- • $\mu(t)$: Expected failures by time t
- • φ : Failure intensity decay parameter
- • θ : Scale parameter
- • t : Time

Interpretation

- • Early in testing: Many failures occur (high $\lambda(t)$)
- • As time progresses: Major faults are fixed, fewer failures occur
- • Logarithmic relationship shows diminishing returns – reliability growth slows over time

Assumptions and Advantages

- Assumptions:
 - Failures follow a Poisson process
 - Each failure fix improves reliability
 - Consistent testing environment
- Advantages:
 - Captures realistic reliability growth
 - Parameters can be estimated from data
 - Simple mathematical form

Limitations and Applications

- Limitations:
 - • Assumes uniform fault reduction per fix
 - • Does not account for new faults introduced
 - • May not fit complex systems
- Applications:
 - • Software testing and debugging
 - • Reliability prediction
 - • Maintenance planning

Example and Summary

- Example:
- Given $\varphi = 0.02$, $\theta = 50$, $t = 100$ hours
- $\mu(100) = (1/50) \ln(1 + 50 \times 0.02 \times 100) = 0.02$
 $\ln(101) \approx 0.092$
- $\rightarrow$ About 9.2% of total expected failures are observed after 100 hours.
- Summary:
- • Type: NHPP reliability growth model
- • Captures decreasing failure rate with time